DT 111 · TOPIC 2

Data Collection and Preprocessing

Sources, sampling, data types, quality, cleaning and preparation

Mkwawa University College of Education

BSc Data Science and Artificial Intelligence

Learning outcomes

- Identify common data sources and collection methods.
- Distinguish populations, samples and common variable types.
- Assess data quality and sampling bias.
- Clean missing, inconsistent, duplicate and invalid data.
- Explain encoding, scaling and other preprocessing steps.



The problem determines the data

The question being investigated determines which evidence is relevant, how it should be collected and whether it can be trusted.

Where data can come from

Primary data

Collected specifically for the current purpose: surveys, interviews, experiments, sensors.

Secondary data

Already exists: census data, administrative records, published datasets, reports.

Internal data

Generated within an organisation: transactions, student records, logs.

External data

Obtained from outside sources: weather, market, demographic or public data.

Population, sample and census

Population

- Entire group of interest
- Example: all DT 111 students
- Often too large or costly to measure fully

Sample

- Subset observed to learn about the population
- Should represent the population well
- Poor selection can create sampling bias

Common variable types

Nominal

Categories without natural order. Example: region.

Ordinal

Categories with order. Example: satisfaction level.

Discrete

Countable numerical values. Example: number of absences.

Continuous

Measurements on a continuum. Example: temperature or weight.

Common forms of data

The form of the data affects storage, cleaning and analysis choices.

Structured

Rows and columns with a defined schema.

Semi-structured

Contains organisation or tags, but not a fixed table structure.

Unstructured

Images, free text, audio, video and other flexible formats.

Five data-quality dimensions

Completeness

Are required values present?

Accuracy

Do values reflect reality?

Consistency

Do formats and meanings agree?

Validity

Do values follow allowed rules and ranges?

Uniqueness

Are unintended duplicates absent?

Example: a small messy dataset

ID	Region	Age	Score
DS014	iringa	21	78
DS015	Iringa	230	65
DS016	Mbeya	20	—
DS016	Mbeya	20	—

“iringa” and “Iringa” should be standardised.

Age 230 violates a plausible range and needs investigation.

The missing score should not be replaced blindly.

The repeated DS016 record may be a duplicate.

A practical cleaning cycle

1

Profile

Inspect types, ranges and missingness

2

Detect

Find errors and inconsistencies

3

Decide

Choose an appropriate action

4

Apply

Clean without overwriting raw data

5

Document

Record what changed and why

Missing values and outliers

Missing values

- May be unknown, unavailable or not applicable
- Possible actions: retain, investigate, remove or impute
- The correct action depends on context

Outliers

- Unusually distant observations
- Can be errors or genuine extreme cases
- Investigate before removing

Duplicates and inconsistent formats

Duplicates

Repeated records may bias counts and summaries.

Categories

“Dar es Salaam”, “dar es salaam” and “DSM” may represent one category.

Units

Centimetres and metres must not be mixed without conversion.

Dates

Consistent date formats reduce ambiguity.

Preprocessing goes beyond cleaning

Cleaning fixes data-quality problems. Preprocessing also transforms data into a form suitable for analysis or modelling.

Common preprocessing operations

Encoding

Represent categories in a computational form.

Scaling

Transform numerical variables so magnitudes are comparable.

Transformation

Apply mathematical changes such as logs or ratios.

Feature selection

Choose variables relevant to the task.

Why scaling can matter

Some algorithms are sensitive to the relative magnitude of numerical features.

Age

18-30

Small numeric range

Annual income

0-60M

Much larger numeric range

After scaling

0-1

Comparable numeric magnitude

Sampling bias can begin before analysis

A perfectly cleaned biased sample is still a biased sample.

Coverage bias

Some parts of the population are missing from the sampling frame.

Non-response bias

People who do not respond differ systematically from those who do.

Convenience sampling

Easy-to-reach participants may not represent the target population.

Raw data and cleaned data

Raw data

- Original unchanged reference
- Supports reproducibility
- Useful for audit and rechecking

Cleaned / processed data

- Contains documented corrections or transformations
- Prepared for analysis
- Should be traceable back to raw data

Good preprocessing practice

- Preserve the original raw data.
- Document assumptions, replacements and exclusions.
- Use domain knowledge to decide whether a value is plausible.
- Do not treat every unusual value as an error.
- Check whether cleaning choices could introduce bias.

Topic 2 key ideas

- Good analysis begins with relevant and trustworthy data.
- Sampling decisions affect representativeness before cleaning begins.
- Data quality includes completeness, accuracy, consistency, validity and uniqueness.
- Cleaning and preprocessing are related but not identical.
- Every important data transformation should be documented and reproducible.